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Machine Learning Guided Design of Doped Lithium Lanthanum Zirconium Oxide (LLZO) for Improved Performance

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ABSTRACT

This paper presents a machine-learning approach to the development of all-solid-state batteries (ASSBs) with doped Lithium Lanthanum Zirconium Oxide (LLZO) electrolytes. LLZO electrolytes are promising solid electrolytes that exhibit high ionic conductivities of approximately $\sim 10^{-3}$ – 10^{-4} S cm⁻¹. However, optimizing the ionic conductivity of LLZO is a multiparametric challenge involving numerous compositional and processing variables. This work employs a data-driven methodology by applying machine learning to a dataset mined from the literature. This approach was used to model the intricate relationships between the compositional and synthesis variables and ionic conductivity. Tree-based and ensemble models, like Decision Tree, Random Forest, Light Gradient Boosting Machine, and CatBoost, are effective for interpreting complex relationships among variables. The models were trained with systematic hyperparameter tuning and rigorous cross-validation to predict ionic conductivity, achieving accuracies from 0.85 to 0.94. Feature importance and partial dependence analyses were used to quantify and visualize variable effects. Across the models, lithium composition and sintering conditions emerged as the most influential factors. Partial dependence plots were used to identify optimal synthesis and compositional ranges. By identifying promising search spaces, this approach can accelerate the design and discovery of high-performance LLZO electrolytes for solid-state batteries.

1 | Introduction

Our environment has suffered significant harm due to the continued reliance on energy from fossil fuels [1]. Therefore, a major goal of modern energy technologies is to enable the adoption of greener and cleaner energy sources. Developing efficient energy-storage solutions is crucial, particularly because of the intermittent and variable nature of renewable energy sources [2, 3]. In 2024, the G7 nations agreed to a new global energy storage target of 1500 GW by 2030 [4]. Similarly, at the 29th

meeting of the Conference of the Parties (COP) to the United Nations Framework Convention on Climate Change (COP29), multiple nations committed to the “Global Energy Storage and Grids Pledge” [5, 6], which also commits signatories to deploying 1500 GW of energy storage by 2030, approximately six times the level achieved in 2022. Reaching this global target by 2030 will require accelerated methods for materials discovery and design.

Materials discovery and development have progressed through four major stages (Figure 1) [7]. For a long time, beginning as

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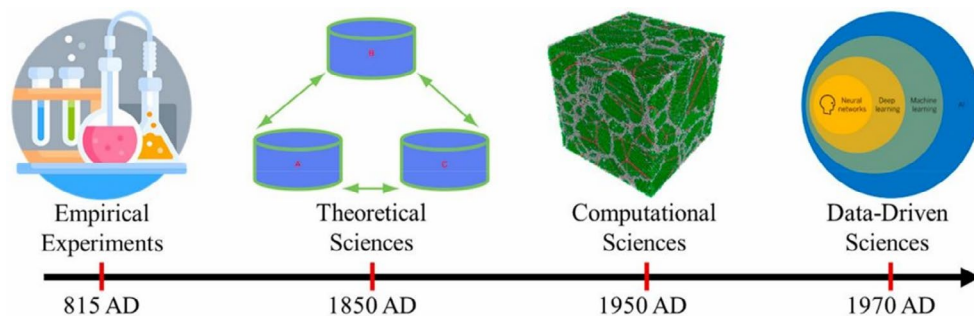


FIGURE 1 | Stages of materials discovery [7].

early as 850 AD, experimental trial-and-error methods were the main approach. In the mid-19th century (~1850 AD), the theory-driven scientific approaches were introduced, using laws and equations to predict materials behavior. By the middle of the 20th century, advances in computing power enabled scientists to design and analyze materials more accurately using simulations. More recently, the large volume of data generated from experiments and simulations have accelerated the growth of data-driven approaches. This approach leverages existing data to make valuable and useful predictions, guide further experiments, reduce costs, and improve the efficiency of materials development [7].

The global energy-storage deficit creates an urgent demand for new technologies that traditional experimental approaches alone cannot address rapidly enough. This requires data-driven methods that can accelerate and guide the research process [8]. Machine learning, in particular, is a powerful tool for accelerating the development of next-generation energy-storage devices. This approach enables the analysis of large datasets to identify material structure–property relationships, accelerate material selection processes, and guide experiments more efficiently [9]. Thus, machine learning can significantly accelerate the design and optimization of new materials.

Among diverse energy storage technologies, lithium-ion batteries (LIBs) have been at the center of battery research due to their high energy density and efficiency in powering portable electronics and electric vehicles [10, 11]. However, safety concerns associated with LIBs exist. Most commercial LIBs utilize flammable liquid electrolytes, which pose risks of thermal runaway, potentially leading to fire or explosion [12, 13]. Solid-state batteries are emerging as promising candidates for next-generation energy storage. This is because they employ solid electrolytes in place of the flammable liquid electrolytes and offer enhanced safety, higher energy density, and improved electrochemical stability [11, 14–16]. Lithium Lanthanum Zirconium Oxide (LLZO) is an inorganic garnet-structured solid electrolyte showing notably high ionic conductivities (10^{-4} – 10^{-3} S/cm) [17–19], nearing that of liquid electrolytes ($\sim 10^{-2}$ S/cm) [18, 20–21]. Improving the ionic conductivity of solid electrolytes is a critical step for optimizing solid-state battery performance and improving energy storage capacity [22].

Machine learning algorithms like Random Forest (RF), Artificial Neural Networks, and Gradient Boosting Machine have been successfully used to predict ionic conductivity from compositional data, achieving high accuracy with R^2 values up

to 0.85 [23]. Active learning techniques using Magpie generated features have been able to predict ionic conductivity from 188 garnet oxide compositions with RF and Artificial Neural Network Models, thereby reducing the experimental effort [2]. The influence of Li site dopants, ionic radius and relative density on ionic conductivity has been examined in another study, which achieved an F1 score of 0.81 and identified V, B and Cr as promising dopants. Machine learning has also been used to identify uncommon synthesis routes for garnet electrolytes through outlier detection models which yielded ROC–AUC (Receiver Operating Curves–Area Under the Curve) values of 0.71–0.72 [24].

The design of high-performance solid electrolytes is a multi-parametric challenge where compositional, structural, and processing variables complexly influence the ionic conductivity. This work explores the use of machine learning to uncover complex relationships in research data with the goal of using these findings to design targeted experiments. This is addressed by applying machine learning models, such as Decision Tree (DT), RF, Light Gradient Boosting Machine (LGBM), and CatBoost (CB), to a dataset mined from literature. Our approach identifies the most influential compositional and synthesis parameters impacting the ionic conductivity and also provides interpretable analyses that capture these relationships. This process identifies quantitative ranges for different doping and compositional variables within which optimal ionic conductivity can be achieved. This study offers practical design guidelines for next-generation solid-state batteries.

2 | Methodology

2.1 | Data Collection and Preprocessing

Data used for machine learning can be obtained from established databases, generated through high-throughput simulations, or mined from literature; many researchers also run large-scale experiments to collect data. In this work, we compiled a database through an extensive literature review. To develop a faster and more cost-effective method for designing and synthesizing doped LLZO, we conducted a literature review to build a database of the synthesis parameters for doped LLZO prepared by the conventional solid-state reaction method, as such a database was not previously available. The dataset was compiled from 55 peer-reviewed papers, selected based on how explicitly the authors documented the synthesis process. This minimized the variability of synthesis conditions and missing

data in the database. The papers had to satisfy the following minimal criteria: (i) used the conventional solid-state reaction method, (ii) reported the processing details, including the calcination time and temperature, sintering time and temperature, the percentage of excess lithium added, the ball milling time, and (iii) stated the total ionic conductivity (not only the bulk or grain value conductivity). These criteria were applied to reduce noise, inconsistencies, and variability arising from differences in experimental procedures. This process resulted in 230 different records. All materials in the database were of LLZO-based compositions, including both doped and undoped systems.

The challenges of data scarcity are well-documented in battery research [9]. The dataset size was constrained because many published papers do not fully report the processing parameters used in their experiments. Furthermore, the dataset size is consistent with several peer-reviewed studies in the field that have developed reliable models using fewer than 200 samples [2, 24–25].

Data preprocessing is an essential step in machine learning because it ensures that the input data are of sufficient quality for extracting useful information [26]. To improve the quality of the data, outlier removal and data scaling were performed as part of preprocessing. In addition, rows with more than three missing values were removed to reduce inconsistencies that could arise from data imputation.

After loading the dataset, the data were cleaned, and missing values were imputed using the Multiple Imputation by Chained Equation (MICE) imputation method. This effective simulation-based method handles missing data by modeling each variable with missing values as a function of the other variables in the dataset. It is a flexible method that fits a distinct predictive model to each data type, making it a preferred approach for data imputation. The method is also advantageous because it preserves the interdependencies between the features [27, 28]. All categorical data were encoded using the one-hot encoding. The one-hot encoding method converts categorical data into a numerical format by creating a new binary column for each unique category. This approach is advantageous because it prevents the machine learning model from assuming a false ranking between the categories and ensures that each is treated as a distinct, equivalent option [29].

The dataset was classified into two classes based on the targets “good ionic conductivity” (1) and “poor ionic conductivity” (0), distinguished by the median ($2.69 \times 10^{-4} \text{ S cm}^{-1}$) (Figure 3e). The threshold also lies well above the documented benchmark of $1 \times 10^{-4} \text{ S cm}^{-1}$ for viable solid-state electrolytes [30, 31]. The chosen benchmark, therefore, allows the model to distinguish comparatively high-performance electrolytes. For small datasets, the median ensures a balanced class distribution to prevent model bias toward larger classes. A similar classification approach has been used in prior work [25, 30]. Figure 2 represents the general approach to the classification task in this study, showing an overview of the process from the data cleaning to the data preprocessing, the ML algorithms used, and model creation after hyperparameter tuning, to model evaluation and selection through to the analysis of the model

predictions via partial dependence plots (PDPs) after fitting the feature importance. We employ a systematic data preparation process that adopts principles from materials science and machine learning. Table 1 carries a description of the feature abbreviations used in this study.

2.2 | Machine Learning Algorithms

The application of machine learning models is instrumental in materials design and discovery by modeling material-property-performance relationships. Tree-based models have emerged as powerful strategies for materials development for LLZO and solid-state batteries. These learning methods, such as the DT [32], RF [2], LGBM [25], and CB [23], have been explored in the study of LLZO and solid-state batteries and have been shown to achieve high performance. The machine learning algorithms described here were employed to train the models used in the analyses presented in this study.

2.2.1 | Decision Tree

DT methods address both regression and classification problems. It makes decisions through rules based on data features forming a binary tree structure [33]. The core nodes in this tree-based model stand for the test on the features, the branches for the output, and the leaves for the classes [31].

2.2.2 | Random Forest

RF Algorithms can also be applied to both classification and regression problems. The RF is an ensemble of DTs and therefore learns based on multiple trees. It is more advanced than the DT algorithm and makes its prediction based on the output from the different trees [32].

2.2.3 | Light Gradient Boosting Machine

This algorithm is based on DTs. A group of weak DT classifiers is used as a boosting method. To arrive at the ultimate decision, the model sums up the individual trees' projections. To get the final predictions, the model totals the forecasts from each tree [34].

2.2.4 | CatBoost

Categorical Boosting (CB) is a Gradient Boosted Decision Tree (GBDT) model. It handles complicated categorical variables and other heterogeneous data. It also addresses problems of overfitting. For tasks involving regression and classification, it works effectively [33].

2.2.5 | AdaBoost

Adaptive Boosting, or AdaBoost (AB), is an ensemble method that enhances weak estimators to produce more reliable

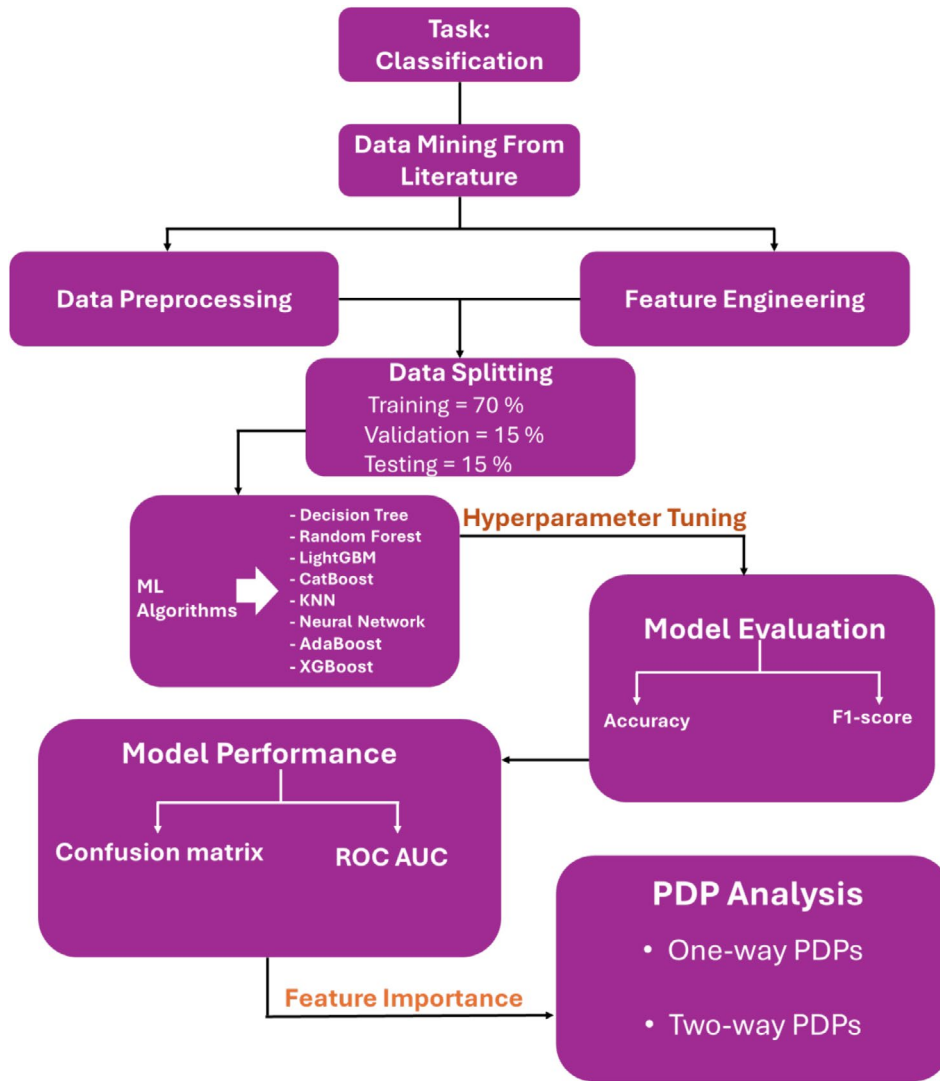


FIGURE 2 | Machine Learning workflow for this study.

TABLE 1 | Feature description.

Feature	Description
Li_comp	Lithium composition
Zr_site_sto	Composition of the Zr site dopant
Zr_site_elnv	Electronegativity of the Zr site dopant
La_site_A	Atomic mass of the La site dopant
Li_site_sto	Composition of the Li site dopant
Li_site_elnv	Electronegativity of the Li site dopant
Zr_comp	Zr composition
Li_dop_Al	Li site dopant aluminium
La_comp	Lanthanum composition
Zr_site_irad	Ionic radius of the Zr site dopant
La_site_sto	Composition of the La site dopant

and accurate predictions. It strengthens weak models by iteratively correcting their errors to build a more accurate predictor [35].

2.3 | Model Training

The predictive models for this binary classification task were developed through a structured process of training, validation, and testing. The targets were labeled 0 (representing “poor” ionic conductivity) and 1 (the preferred target, representing “good” ionic conductivity). The data was divided into a training set (70%), a validation set (15%), and a testing set (15%). To ensure a proportional distribution of the binary target classes (0 and 1) across the three subsets, a stratified splitting strategy was implemented. This method guarantees that each subset accurately reflects the entire dataset. Stratification reduces bias and ensures that the model is exposed to a representative proportion of each class during training, evaluation, and testing that accurately reflects the true data distribution [36].

2.4 | Model Evaluation Metrics

Relying on one evaluation metric can provide an incomplete perspective on model performance; therefore, it is essential to use multiple metrics. Collectively, diverse metrics offer a thorough

TABLE 2 | Feature categories.

Material stoichiometry	Dopant related features	Synthesis parameters
Li, La, Zr composition	Electronegativities	Calcination parameters
Li, La, Zr site dopant stoichiometry	Ionic radii	Ball milling parameters
	Atomic mass	Sintering parameters
		Pelletizing pressure

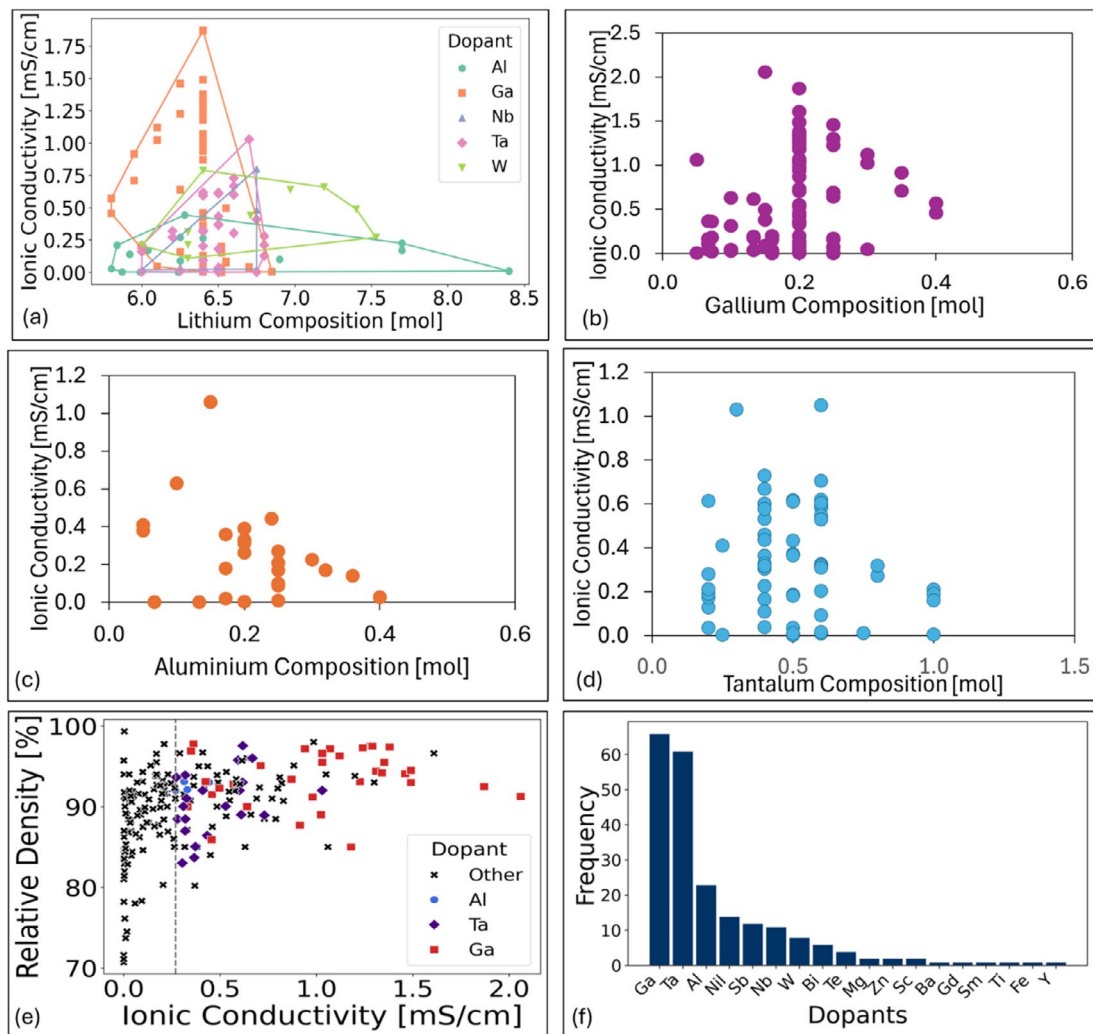


FIGURE 3 | Data exploration showing plots of ionic conductivity versus (a) lithium composition, (b) gallium composition, (c) aluminium composition, (d) tantalum composition; also plots of (e) relative density versus ionic conductivity and (f) the frequency of dopants in the dataset. These plots reveal compositional trends and their influence on ionic conductivity.

evaluation of the model's performance. In evaluating the models, the following metrics were used to gain an understanding of their ability to distinguish between the classes.

Accuracy: This refers to the proportion of correct predictions to the total number of predictions [37]. The accuracy is computed from Equation (1) as follows:

$$\text{Accuracy} = \frac{\text{Number of correctly predicted}}{\text{Total number of predictions}} \quad (1)$$

Precision: This indicates the number of positive outputs that the model correctly predicted out of the model's positive predictions [37], calculated as shown in Equation (2).

$$\text{Precision} = \frac{\text{True positives}}{\text{Total predicted positives}} = \frac{\text{True positives}}{\text{True positives} + \text{False positives}} \quad (2)$$

Recall: This tells the proportion of positive outcomes predicted by the model that were positive [37]. Recall is computed from Equation (3).

$$\text{Recall} = \frac{\text{True positives}}{\text{Total actual positives}} = \frac{\text{True positives}}{\text{True positives} + \text{False negatives}} \quad (3)$$

F1 Score: This combines precision and recall into a single number as shown in Equation (4). The F1 score gives equal weight to

both precision and recall; a higher score indicates better model performance [37].

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Receiver Operating Characteristic (ROC) Curve: The Receiver Operating Characteristic (ROC) curve is an established method for evaluating the effectiveness of binary classification models. It presents the true positive rate (Equation 5) against the false positive rate (Equation 6) [37], allowing us to assess how well the model can differentiate between the two classes. A key metric derived from the ROC curve is the area under the curve (AUC), which quantifies the model's proficiency in class distinction [24, 32]. This plot shows the performance of the classifier over all possible thresholds [37].

$$\text{True positive rate (TPR)} = \frac{\text{True positives}}{\text{True positives} + \text{False negatives}} \quad (5)$$

$$\text{False positive rate (FPR)} = \frac{\text{True positives}}{\text{False positives} + \text{True negatives}} \quad (6)$$

2.5 | Feature Importance

Feature importance quantifies the relative contribution of each feature in a model. It indicates the degree to which each feature informs the model's prediction [37]. Analyzing feature importance provides deeper insight into the data and allows focus to be directed toward the predictors with the greatest impact for further investigation [32].

2.6 | Partial Dependence Plots

PDPs are useful in interpreting the predictions of machine learning models. They show the dependence between the target value and the features of interest. This is usually between a small set of features, which is typically chosen among the most important features. These plots show complex or linear relationships between the features and the target values, as well as the response of the target when one input variable is modified. PDPs are model-agnostic and can therefore be used to interpret the predictions of different machine learning models [38, 39]. The most important features are obtained via the feature importance obtained from each of the trained models.

One-way PDPs show how the prediction of a model is dependent on one feature in the model. PDPs with two features, called "Two-way partial dependence plots" (Two-way PDPs), enable us to visualize the interactions between them, illustrating how one feature depends on another to achieve the desired target.

3 | Results and Discussion

3.1 | Description of Experimental Data

Table 2 shows the different categories of features contained in the database spanning over material stoichiometry, dopant-related features, and processing parameters. An exploratory

statistical analysis of the data is shown in Figure 3a–e. Figure 3a reveals the relationship between lithium composition and ionic conductivity grouped by different dopants.

Close observation shows that the gallium-doped LLZO demonstrates considerably higher ionic conductivity than the other dopants represented in Figure 3a. There is also a significant cluster of the ionic conductivity values around 6.4 mol lithium composition, reflecting the importance of this composition to the ionic conductivity of these materials. However, the ionic conductivity values vary widely within this composition, indicating that factors besides the composition also play a considerable role in influencing the ionic conductivity values.

The data collected included the relative density of the materials, which can be seen to have a high correlation with the ionic conductivity from the trend in Figure 3e. During the model training and analysis, the relative density was not included, as only synthesis and pre-synthesis parameters were used; the relative density can only be calculated after the material has been fully prepared. In Figure 3f, the various dopants present in the dataset are shown along with their frequency of occurrence. As shown in Figure 3f, the database contains the features of 17 distinct dopants, with gallium (Ga), tantalum (Ta), and aluminium (Al) having the highest occurrences. Figure 3b–d presents an overview of the distribution of ionic conductivity with the composition (mol) of these dopants. A comparison of these plots shows that the Ga dopant (Figure 3b) achieves a significantly higher ionic conductivity than the others.

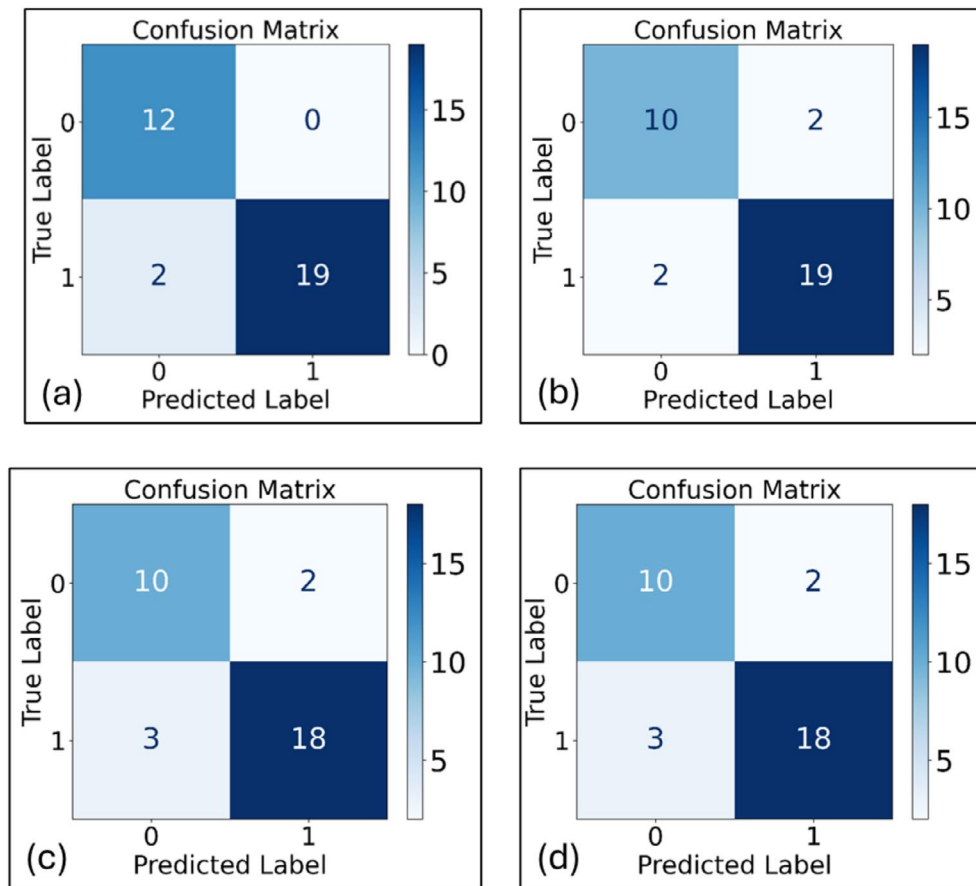
When working with datasets that contain several variables, exploring the data to discover useful knowledge can be challenging. Unlike the traditional statistical approaches, machine learning is increasingly being used to extract and interpret the relationships between different features. These relationships are sometimes simple and linear, but at other times, they can be more complex and difficult to detect using conventional statistical approaches.

3.2 | Model Validation

GridSearchCV was employed to optimize model performance by searching for the best parameters. GridSearchCV is a method of hyperparameter tuning for machine learning models, which performs a search over a predefined grid of hyperparameter values defined based on a series of exploratory experiments. It then performs cross-validation using each possible combination of the parameters; after this, the combination with the best performance is chosen to train the models [29]. In this study, GridSearchCV was performed with a five-fold validation. For each hyperparameter combination (Table S1), models were trained on four folds and validated on one-fold. The weighted F1 score was used as the main metric to ensure a balanced evaluation of precision and recall. A total of eight classification algorithms were examined; following hyperparameter tuning, validation, and testing, the top four models were selected. Based on the model evaluation summarized in Table 3, the DT, RF, LGBM, and CB Models demonstrated high values of all key metrics—accuracy, F1 score, precision, recall, and ROC-AUC scores (Table 3). These models, with accuracies, precision, recall,

TABLE 3 | Model validation.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	ROC AUC (%)
Decision Tree	93.94	94.81	93.94	94.02	95.24
Random Forest	87.88	87.88	87.88	87.88	91.07
LightGBM	84.85	85.24	84.85	84.97	87.7
CatBoost	84.85	85.24	84.85	84.97	92.46
XGBoost	81.82	82.91	81.82	82.06	91.67
Neural Network	72.73	73.29	72.73	72.94	85.32
AdaBoost	72.73	74.85	72.73	73.19	78.77
KNN	66.67	73.68	66.67	67.1	75.6

**FIGURE 4** | Confusion matrix of (a) Decision Tree, (b) Random Forest, (c) LightGBM, and (d) CatBoost models; illustrating each model's performance in distinguishing between the target classes.

and F1 scores above 84%, and ROC-AUC values above 87% were used as final models for subsequent analysis and to extract meaningful information from the data.

Figure 4 presents the confusion matrices of the fitted models. The confusion matrix enables the evaluation of models by presenting a tabular layout comparing the predicted class labels against the actual labels over the test data. With this, the pairs of classes that the model confuses are easily visualized [40]. The confusion matrices in Figure 4a–d provide a detailed summary of the models' performance on the test set. By comparing the

four confusion matrices, the strength of each model in distinguishing between the two classes is demonstrated by matching the true and predicted values. The models consistently demonstrate a strong ability to identify the classes correctly.

The ROC curves and AUC scores were used to validate the selected classification models, which showed good discriminating power with AUC scores of 0.88, 0.91, 0.92, and 0.95 for LightGBM, RF, CB, and DT, respectively (Figure 5). For perfect classification, the AUC score will be 1, while for purely random classification, it will be 0.5 [29]. The analysis of the ROC curves

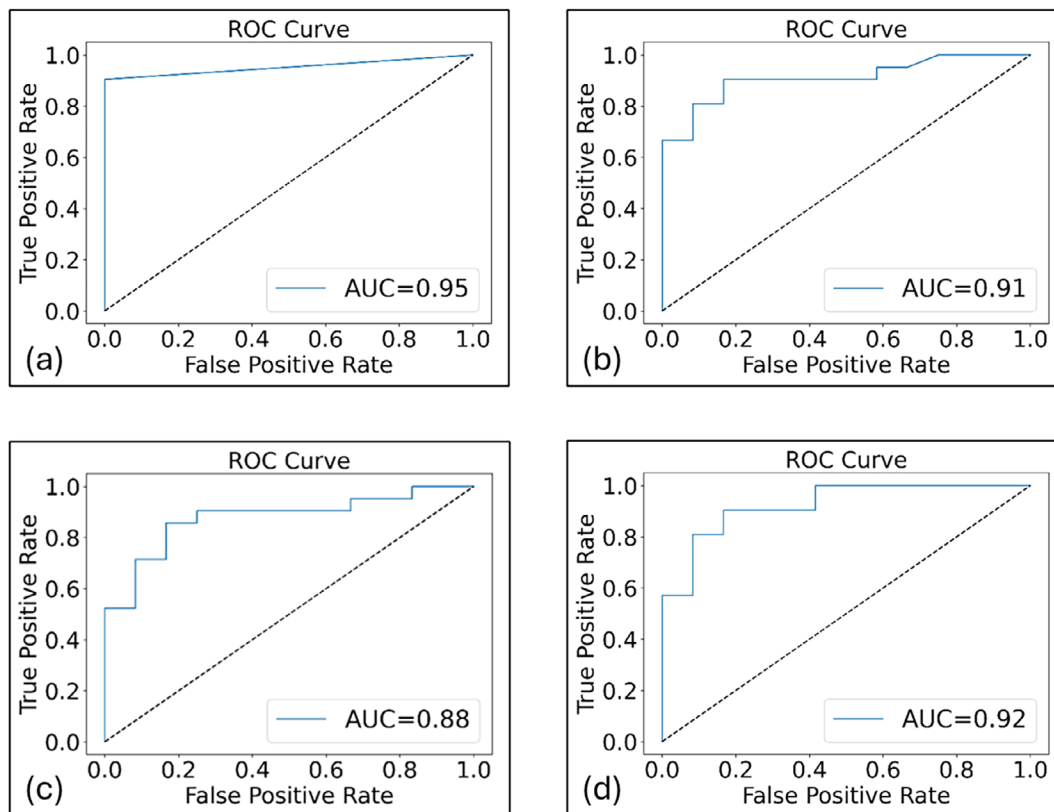


FIGURE 5 | Receiver Operating Curves [ROC] performance curves of (a) Decision Tree (Area Under the Curve [AUC]=0.95), (b) Random Forest (AUC=0.91), (c) Light Gradient Boosting Machine (AUC=0.88), and (d) CatBoost (AUC=0.92) models; the AUC values indicating the strong predictive accuracy and discriminative capability of the models.

and the AUC scores shows that all four models are effective and have a strong classification performance. These models are ensembles of DTs, and this shows how important this kind of architecture is to the problem at hand [25].

3.3 | Model Evaluation and Interpretation

3.3.1 | Feature Importance

Beyond obtaining a high accuracy, it is important to understand the model predictions by identifying the key drivers behind them. The optimized models are used to identify the key features influencing the ionic conductivity of the electrolyte by leveraging the built-in feature importance of the fitted models. Furthermore, the relationship between ionic conductivity and these key features is examined through the feature importance and PDPs. Different machine learning models have unique architectures, which can lead to a variation in their interpretation of feature importance. By comparing the findings from the DT, RF, LGBM, and CB models, we discover the most important processing conditions (Figure 6).

A very significant finding of this comparative analysis is the high degree of stability and substantial consensus in the features identified as most important. Despite using four different models, the set of most important features remained largely the same. Examining the top 15 features from each model revealed a significant overlap with at least 10 features reappearing across all models. This indicates a high degree of predictability

regardless of the algorithm. The implication is that the identified features are universally important, while their ranks reflect the model biases; thus, they represent the most important drivers for predicting the target variable.

The consensus across all four models, from a single DT to advanced boosting models, identifies the Lithium composition as the most powerful predictor in the dataset. This indicates a strong and fundamental relationship with the target variable, independent of model architecture. The sintering conditions of temperature and time rank as the next most important features across the models.

The ranking of the sintering conditions as the second most important feature suggests that this condition is very influential in achieving optimal ionic conductivity. The sintering process has a high impact on the densification of the pellets and hence the relative density of the material, which, as we have shown in Figure 2 contributes highly to the ionic conductivity of these materials; hence optimal sintering conditions will lead to higher densification, higher relative density, and in turn an improved ionic conductivity of the material [41].

The Lithium composition is closely linked to the number of vacancies available for Li-ion mobility. These vacancies are created by removing Lithium ions through the process of doping. This underscores the significance of both Lithium site and zirconium site doping as highlighted by the model feature importances. Understanding these relationships is crucial for optimizing the ionic conductivity. The effects of Li site dopants like Al and Ga

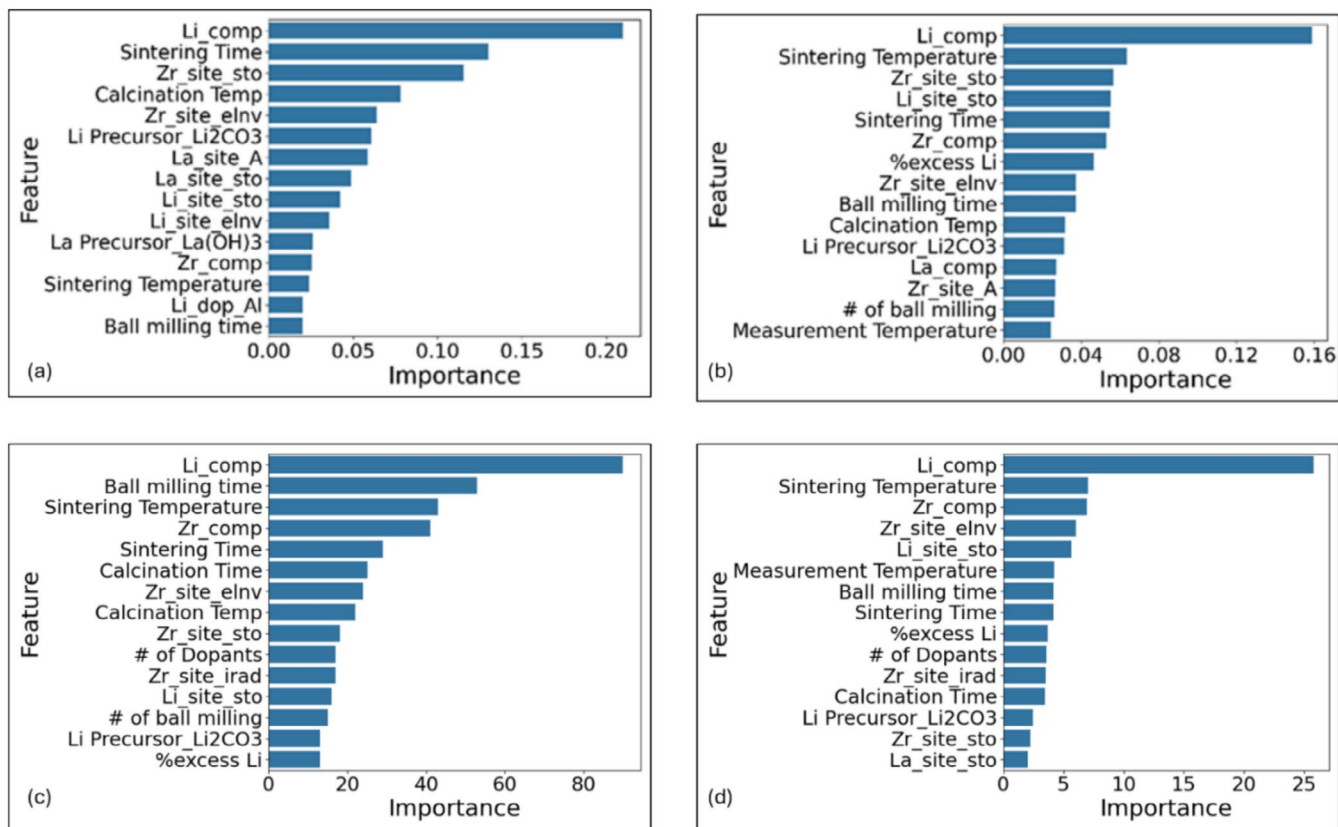


FIGURE 6 | Comparative analysis of feature influence on ionic conductivity based on the (a) Decision Tree, (b) Random Forest, (c) Light Gradient Boosting Machine, (d) CatBoost models. *Note:* The symbol ‘#’ is used in this figure to denote “number.”

in stabilizing the cubic phase for high room temperature ionic conductivity in garnet-structured LLZO electrolyte have been established [42–44].

Similarly, the models attribute high importance to the composition of the dopant on the zirconium site as well as the zirconium composition. This indicates that zirconium site doping plays a crucial role in enhancing the conductivity of doped LLZO electrolyte. Zirconium site dopants like Ta and Nb have been associated with improved ionic conductivity [45, 46]. Additionally, the electronegativity of the zirconium site dopant emerges as another important feature identified by the models, further reinforcing the importance of the zirconium site doping strategy in achieving high conductivity and stability in LLZO electrolyte [45, 46].

Carefully controlling the milling conditions is vital for tailoring the microstructure and maximizing the performance of LLZO for solid-state battery applications [47, 48]. Nguyen and Park [48] elucidated the importance of the second ball milling step for obtaining smaller-sized particles for LLZO, stating that larger-sized particles lead to higher porosity when sintered at high energy, which could encourage the growth of dendrites, and they cannot be sintered fully at low energy.

3.3.2 | Partial Dependence Plots

PDPs were used to interpret the predictions of the classification models and to assess the impact of individual features on the target ionic conductivity class. This analysis was used to identify

specific feature ranges and combinations associated with the highest probability of an optimal outcome.

Figure 7 shows the PDPs for the four selected models, focusing on the five most important features, to evaluate the conditions most likely to produce optimal ionic conductivity. Each feature exhibits a different trend in the plot, showing its unique effect on the models' predictions.

As noted above, the composition of lithium is the most critical feature identified by the selected models. All models suggest that the optimal lithium content falls within the range of 6.1–6.4 mol, with 6.4 mol being recognized as the peak value by two of the models (see Figure 7a,f,k,p). The PDPs demonstrate an improvement in the probability of achieving higher ionic conductivity as lithium composition increases from 6.1 mol, peaking at approximately 6.4 mol, beyond which it starts to decline [49]. The observation agrees with existing literature on lithium composition, which indicates that this specific range represents the optimal lithium content, as the slight lithium deficiency increases the number of vacant sites available for Li^+ mobility. A higher vacancy concentration could lead to the collapse of conduction pathways within the material [50, 51].

In Figure 7b, the analysis from the DT model indicates that the optimal sintering time range for achieving desirable conductivity is 18–20 h; beyond this interval, the probability of decline in ionic conductivity increases. Complimentarily, the RF Model corroborates this finding, suggesting 20 h as the optimal time value.

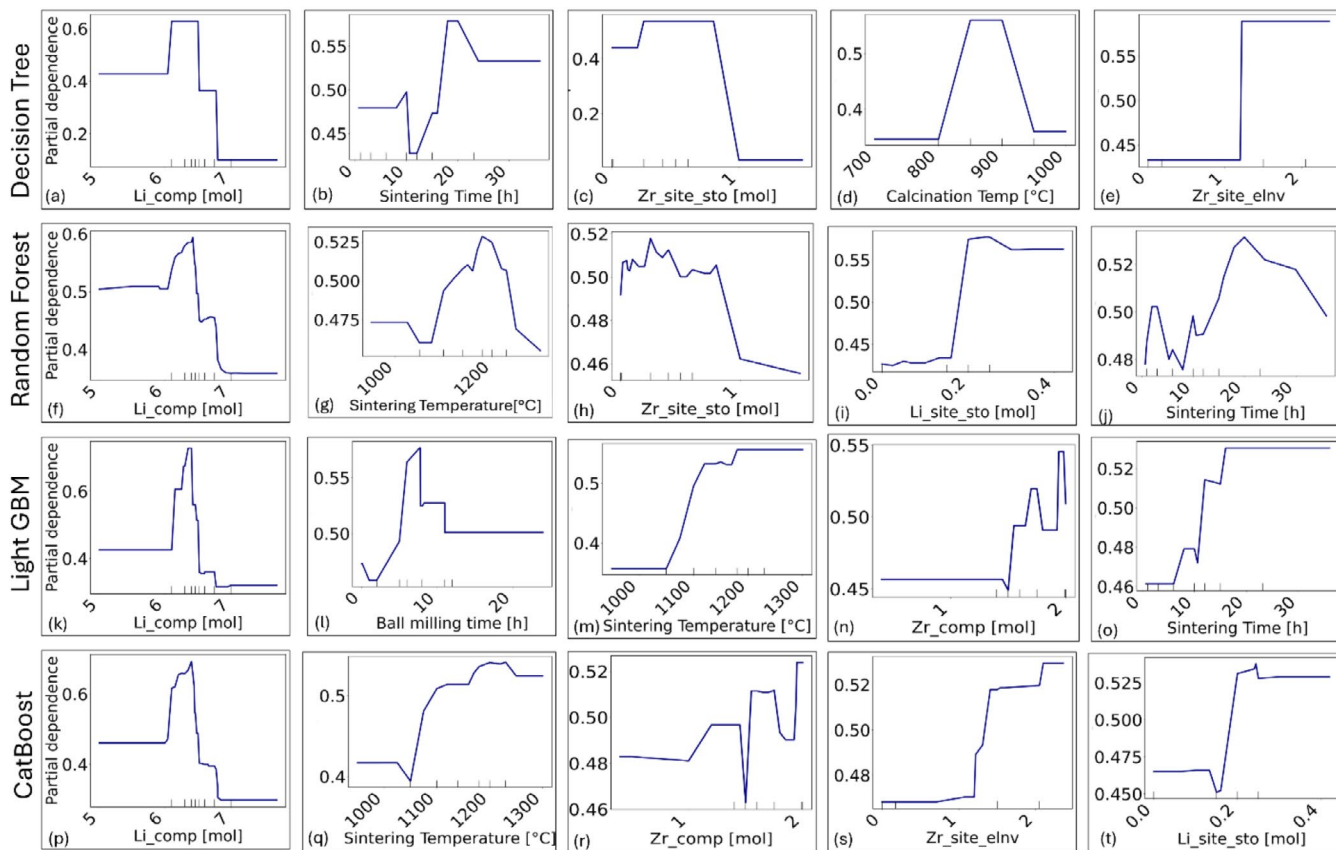


FIGURE 7 | Partial dependence plots illustrating the relationships between the top five most important features and the predicted outcomes for (a–e) Decision Tree, (f–j) Random Forest, (k–o) Light Gradient Boosting Machine, and (p–t) CatBoost models.

The PDPs illustrating the relationship between sintering temperatures and ionic conductivity (Figure 7g,m,q) reveal a consistent range of optimal values peaking at approximately 1200°C for optimal output. These observations align well with the existing literature, where the range for cubic structure formation is suggested to be between 1000°C–1200°C [52, 53].

Additionally, the effects of calcination temperature, illustrated in Figure 7d, reveal a bell-shaped curve. The curve indicates that as the calcination temperature is increased, starting from 800°C, there is a progressive increase in ionic conductivity. The peak is observed in the temperature range of 850°C–900°C. However, surpassing this optimal range results in a slight decrease in the probability of achieving the desired ionic conductivity, suggesting that excessively high calcination temperatures may adversely affect material performance [43, 54].

Furthermore, analysis of the zirconium site dopant presented in Figure 7c,h indicates that an optimal doping range lies between 0.18 and 0.8 mol, with Figure 7h showing a peak performance at dopant composition of 0.25 mol from the RF model. The optimal zirconium composition and electronegativity of the zirconium site dopants are also shown by the PDPs (Figure 7e,n,r,s), further emphasizing the importance of zirconium site doping in the performance of LLZO electrolyte. Optimal values of 1.9 mol by the LGBM model for zirconium composition indicate the combination of zirconium site doping with Li or La site doping for optimal Li vacancy generation and ionic conductivity, while 1.5 mol identified by the CB model most likely indicates optimal

composition for Zr-site doping only. The electronegativity is shown to yield maximum probabilities for electronegativities >1.22. The stoichiometry of the Li-site dopant, as shown by the PDPs (Figure 7i,t), shows optimum values between 0.2 and 0.3 mol with maximum probability at 0.24 mol. This specifies ranges within which Li-site doping is expected to give optimal ionic conductivity results. These findings are consistent with previous studies indicating that both lithium and zirconium site doping strategies can effectively generate the necessary lithium vacancy concentration [19]. However, achieving the same vacancy level requires a higher concentration of zirconium site dopants per formula unit compared to lithium site dopants [44].

The effect of the pelletization pressure on the ionic conductivity could not be determined using the broad dataset due to the limited availability of reported data in the published papers. To address this, a subset of data containing all available values extracted from the literature was created. The impact was then assessed through PDPs generated from a RF model, which demonstrated an accuracy of 90% on the test data.

As shown in Figure 8, a consistently positive relationship is observed between the pressure and the ionic conductivity. Notably, at approximately 200 MPa (Figure 8a), a peak is observed that plateaus, thereafter, suggesting that increments in pressure beyond this point contribute little to no enhancement in the predicted ionic conductivity. Furthermore, the entire curve is positioned above the 0.5 probability, indicating that while the pelletization pressure does not serve as a primary

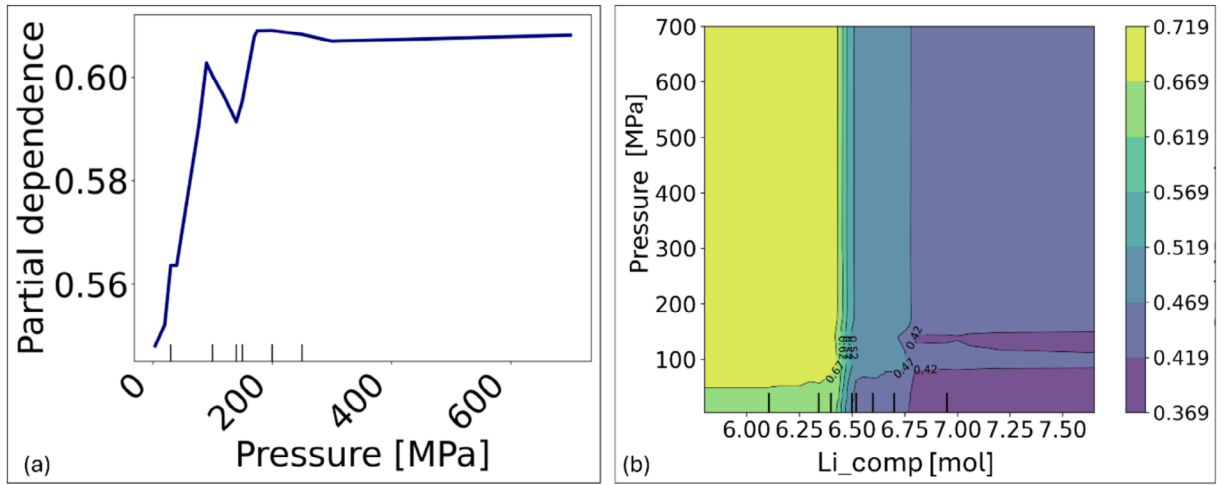


FIGURE 8 | (a) One-way and (b) two-way partial dependence plots showing the effect of the pelletizing pressure on the ionic conductivity. *Note:* The color scale indicates the probability of achieving the desired outcome.

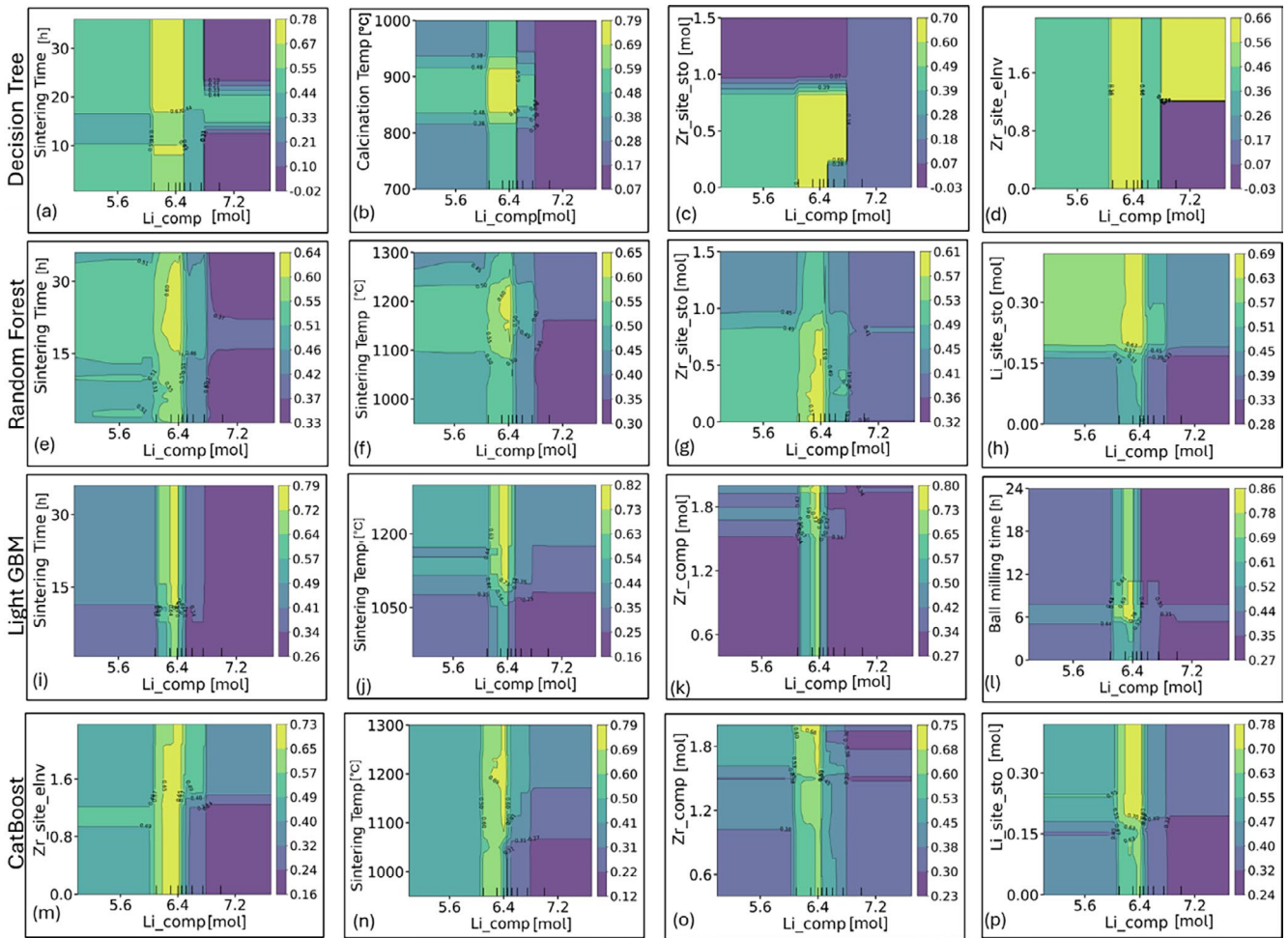


FIGURE 9 | Two-way partial dependence plots depicting the interaction effects between the top-ranked features and lithium Composition from (a–d) Decision Tree (e–h) Random Forest, (i–l) Light Gradient Boosting Machine, and (m–p) CatBoost Models. *Note:* The color scale indicates the probability of achieving the desired outcome.

determinant of the outcome, it possesses the potential to optimize an already favorable outcome. Figure 8b shows the probability of achieving an optimal outcome for a wide range of pressures (~50–700 MPa).

Figure 9 presents the PDPs for two features, allowing us to visualize how multiple features affect the target class of ionic conductivity. The model analyses have highlighted lithium composition as the primary influencing factor. Consequently, the

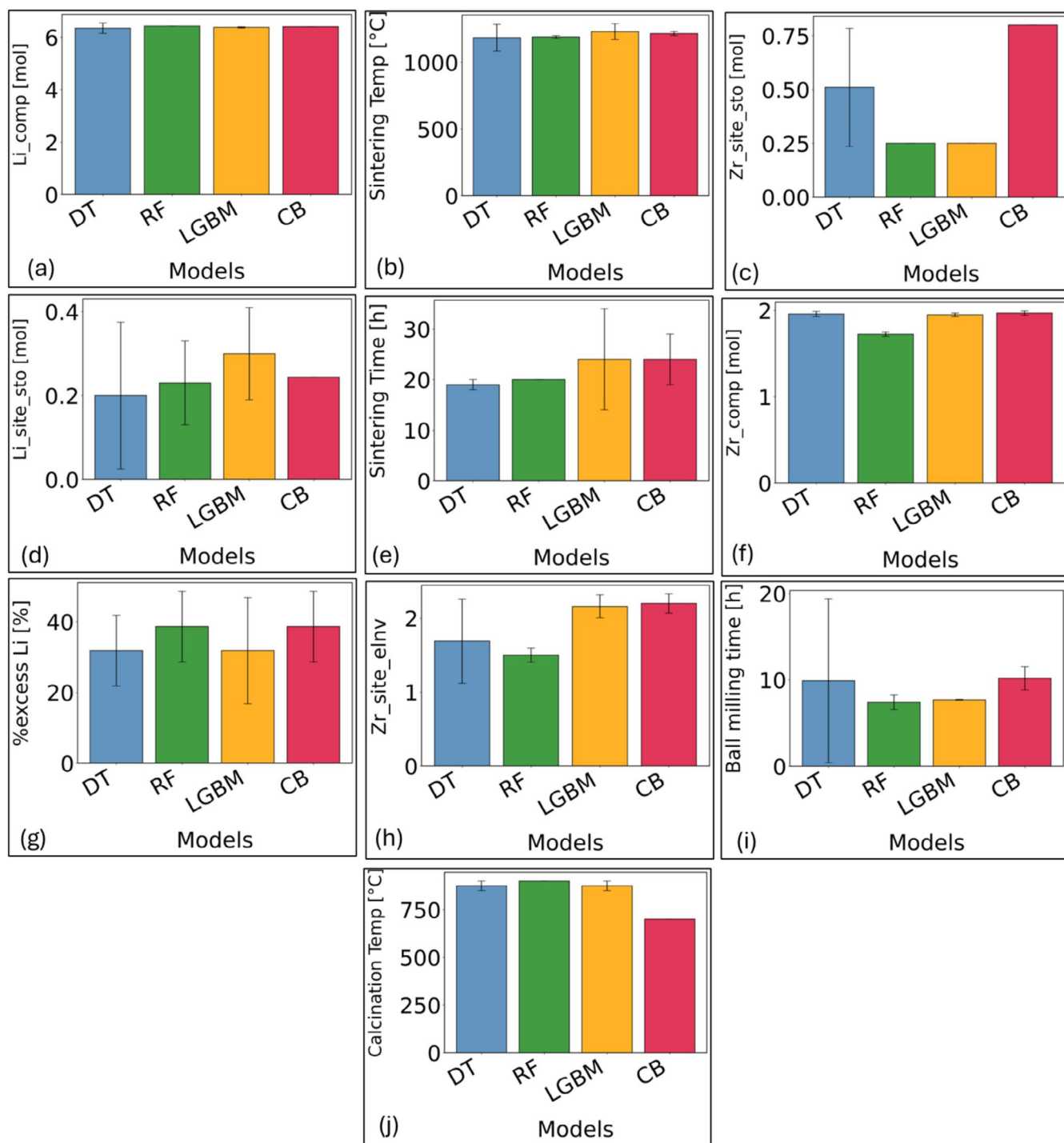


FIGURE 10 | Summary bar plots depicting average PDP values for (a) lithium composition (b) sintering temperature (c) zirconium site stoichiometry (d) lithium site stoichiometry (e) sintering time (f) zirconium composition (g) excess Lithium % (h) zirconium site electronegativity (i) ball milling time, and (j) calcination temperature, from the Decision Tree (DT), Random Forest (RF), Light Gradient Boosting Machine (LGBM), and CatBoost (CB) models.

two-way PDP plots illustrate the interactions between lithium composition and other features identified by the four models.

The implementation of the two-way PDPs enables a clear visualization of the operational domains for the features illustrated in Figure 9a–p. The yellow regions within these plots represent the zones where the likelihood of attaining the desired conductivity is maximized. This enhances understanding of the operational

parameters and gives valuable insights into optimizing performance within the identified regions. The plots reveal that a high probability of the target outcome is consistently achieved when the lithium composition is approximately 6.4 mol. This optimal region is further constrained by the values of the other features. However, the DT model predicts an expansion of the optimal lithium composition range for zirconium compositions between 0.3 and 0.8 mol and gives an optimal lithium composition range

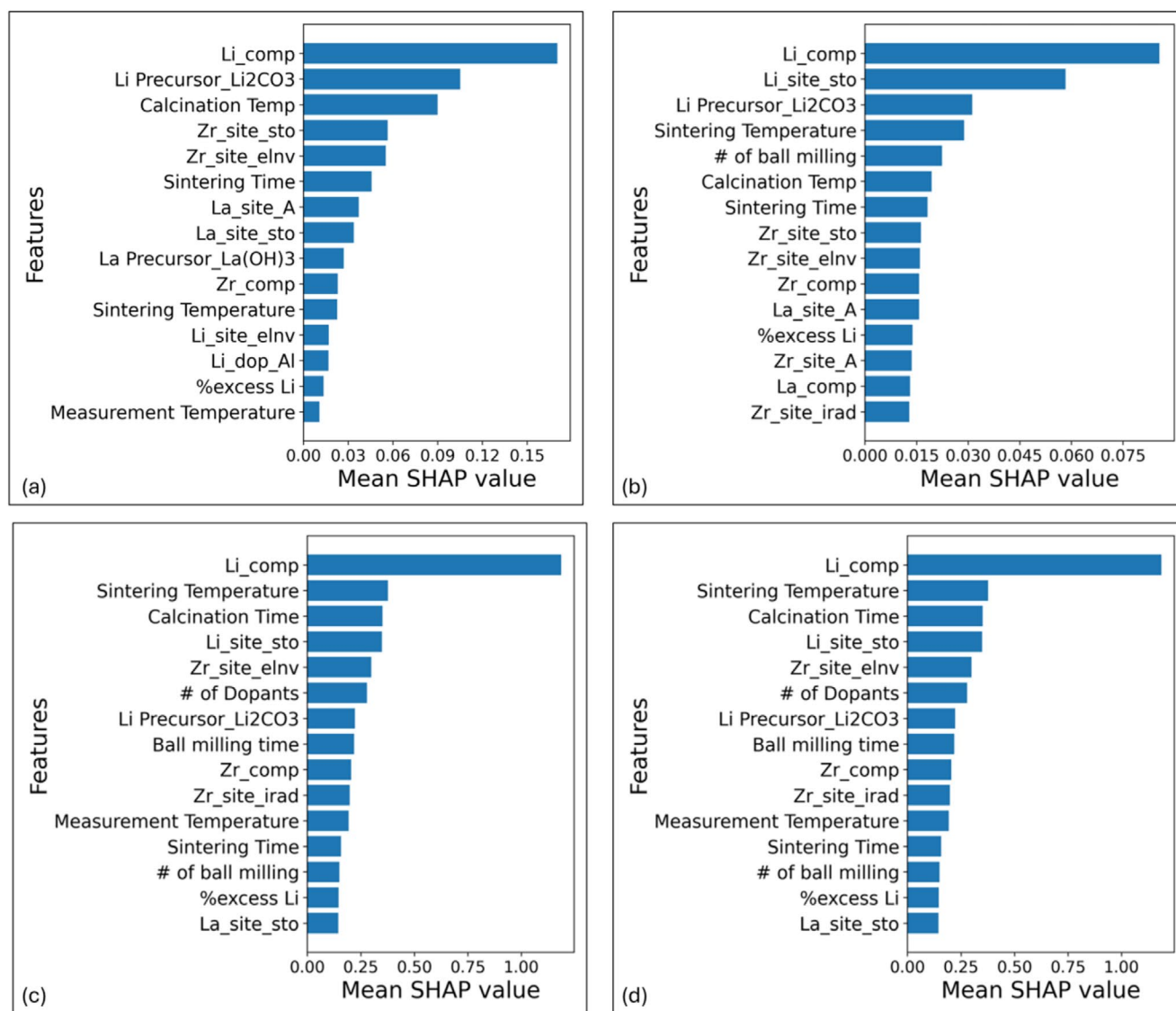


FIGURE 11 | SHapley Additive exPlanations (SHAP) feature importance plots for the top four performing machine learning models: (a) Decision Tree, (b) Random Forest, (c) LightGBM, and (d) CatBoost. *Note:* The symbol “#” is used in this figure to denote “number.”

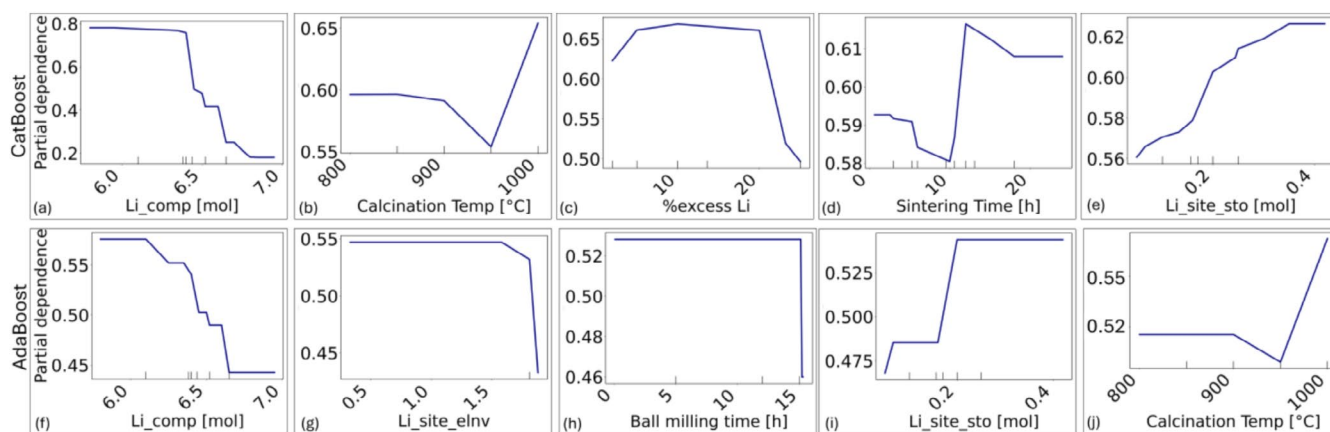


FIGURE 12 | Partial dependence plots illustrating the relationships between the top five most important features and the predicted outcomes based on the Ga-doped LLZO dataset (a–e) CatBoost, and (f–j) AdaBoost.

of 7.2 mol when the zirconium site electronegativity is above 1 (Figure 9c,d). This suggests that modifications at the zirconium site could stabilize a higher optimal lithium composition.

The summary bar plots for the model predictions are shown in Figure 10 for the DT, RF, LGBM, and CB models. The models demonstrated high agreement, identifying similar optimal

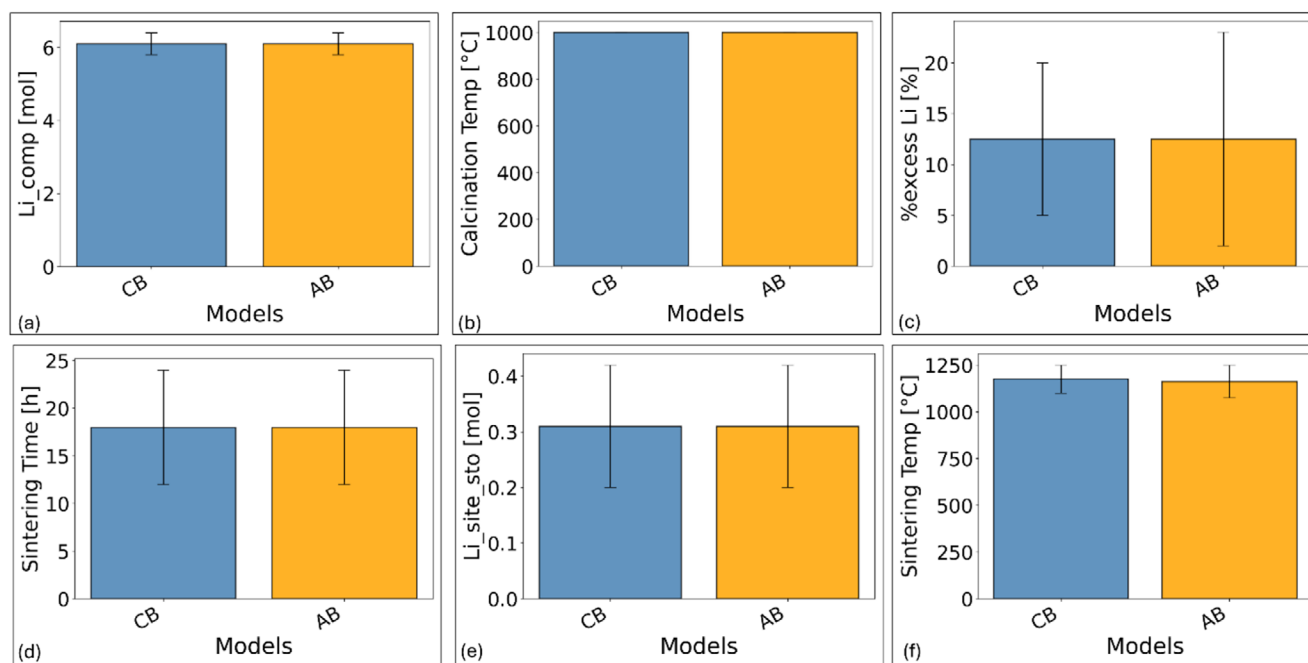


FIGURE 13 | Summary bar plots showing average PDP values of the Ga-doped dataset for (a) lithium composition, (b) calcination temperature, (c) % excess Lithium, (d) sintering time, (e) lithium site stoichiometry, and (f) sintering temperature, from the CatBoost (CB) and AdaBoost (AB) models.

ranges for both lithium composition and sintering temperature. The remaining features exhibited only slight variations, with at least two models in strong agreement. The only exception was zirconium site stoichiometry, which showed wide variations in the predicted ranges. This high degree of agreement across most of the features shows the reliability of the models.

A Shapley Additive exPlanations (SHAP) analysis was performed to further validate the structure–property relationships identified in this study. In agreement with the model-derived feature importance results, lithium composition is consistently identified as the primary driving parameter across all models. As illustrated in Figure 11a–d, when considering the 10 key features highlighted in Figure 10, five features consistently appear in the top SHAP rankings across all four models; three recur in three models, and two are present in at least two models. This recurrence demonstrates that the principal parameters influencing the ionic conductivity are identified irrespective of the interpretability approach employed. While SHAP rankings are not expected to perfectly mirror the model feature importance, the agreement observed here provides independent confirmation of the model interpretations. This reinforces the reliability of these findings for guiding material design and synthesis.

3.3.3 | Gallium Doped LLZO

The data exploration presented in Figure 3 revealed that gallium was the most prevalent dopant in the dataset and exhibited the highest ionic conductivities. A subset comprising 88 entries for gallium-doped LLZO, including co-doped and triple-doped samples, was extracted to train machine learning models. CB and AB models showed the highest accuracies in interpreting this dataset, with accuracies of 92% for both

models. These models were used to examine how the processing and compositional parameters affect the ionic conductivity of gallium-doped LLZO. While most parameters retained similar values for the key features discussed in Section 3.3, there were variations observed in the calcination temperature and sintering time.

The PDPs for this subset are shown in Figure 12 with the summary bar plots for the model predictions shown in Figure 13. Notably, the calcination temperature for this data set is predicted to be 1000°C, which differs from the 850°C–900°C predicted for the general dataset. Also, a significant peak in sintering temperature at 12 h differs from the average of 18–24 h in Figure 10. These observations indicate that more dopant-specific data could explain the relationship between the dopant and the electrolyte performance.

3.4 | Implications for the Design of Solid-State Electrolytes and Batteries

The results from these models give us a broad overview of the effects of the different doping strategies, the Li, La, and Zr site doping on the LLZO performance, and, with the use of the PDPs, define the compositional spaces within which these doping strategies can give optimal output. The effect of the processing parameters is also elucidated in a way that outlines the optimal parameters for each process. The ability of the models to identify the subtle dependencies among the features emphasizes the effectiveness of machine learning and its potential in advancing materials development. This capability can lead to reduced costs by decreasing the number of experiments required, thereby saving researchers both time and resources. Beyond that, the flexibility of this approach allows it to be applied across various processes, enabling investigators to leverage the growing global database to

inform future experiments and designs. Importantly, the findings of this study will serve as a framework to guide subsequent experimental efforts aimed at enhancing the ionic conductivity and overall performance of LLZO-based solid-state batteries.

With an expanded dataset that incorporates a broader range of dopant types and more detailed processing conditions, future research can more precisely uncover the influence of specific dopants rather than their crystallographic site. Such improvements could also lead to enhanced accuracy on the trained models. Additionally, further exploration of alternative processing methods is necessary. Processing methods such as the sol-gel technique or spark plasma sintering can be explored, with systematic comparative studies that may yield valuable insights into the factors governing ionic conductivity.

The dataset resulting from this study has been made publicly available on GitHub (<https://github.com/soboyejoresearch/Battery-Research>) to facilitate further research. Future work will include expanding this database to encompass different processing methods for LLZO batteries. Additionally, experimental investigations will be conducted to further validate these findings and gain deeper insights into the factors that influence the ionic conductivity and overall performance of LLZO batteries.

4 | Conclusions

This study presents the results of a combined data science and materials science study of the effects of compositional and processing parameters on the ionic conductivity of LLZO solid-state electrolytes. The research successfully integrates machine learning techniques with critical domain knowledge to gain insights and make predictions about optimal processing parameters and doping strategies for the synthesis of doped LLZO. This presents a method that can be used to accelerate the development of high-performance solid-state batteries for energy storage applications. Salient conclusions arising from this study are itemized below:

- Optimal processing parameters were identified using machine learning models.
- The PDPs show how the processing and compositional parameters interact to achieve optimal ionic conductivity values.
- Relationships between processing and compositional parameters and the ionic conductivity were identified. The lithium composition is the most influential feature according to different models.
- Sintering parameters significantly impacted the ionic conductivity of doped LLZO electrolytes in most of the models, which can be attributed to their role in the densification of the electrolyte and the consequential improvement of the relative density.
- The findings and methodology in this study can be adapted to guide the synthesis of other materials.
- The parameters identified as important are relevant for developing flexible and scalable LLZO electrolytes.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data is publicly available on the repository of the corresponding author on GitHub with the link: <https://github.com/soboyejoresearch/Battery-Research>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Final hyperparameter search space used in GridSearchCV.